KSA Office IT Support & Infrastructure Management

Phase 07: IT Support Operations

Afaq Business Solutions Co.

System Performance Troubleshooting Standard Operating Procedure (SOP)

1. Purpose

Provide a structured process for diagnosing workstation performance problems using safe, evidence-based Linux troubleshooting methods.

2. Scope

Applies to IT Support personnel investigating slow applications, high resource utilization, and related workstation performance complaints.

3. Performance Troubleshooting Workflow

- Document the user's reported symptoms.
- Review current system load.
- Check memory and swap utilization.
- Identify processes with elevated CPU consumption.
- Identify processes with elevated memory consumption.
- Check whether disk capacity could contribute to the problem.
- Analyze the collected evidence before making changes.
- Apply only safe and authorized remediation.
- Verify the result and document the outcome.

4. System Load Analysis

- Use uptime to obtain a quick view of system load averages.
- Consider load together with CPU, memory, and process information.

5. Memory Analysis

- Use **free -h** to review total, used, available, and swap memory.
- Correlate memory pressure with process-level memory usage.

6. Process Analysis

- Use **ps aux --sort=-%cpu** to identify high CPU utilization.
- Use **ps aux --sort=-%mem** to identify high memory utilization.
- Do not terminate an unknown or critical process solely because it appears high in a process listing.

7. Disk Capacity Check

- Use **df -h** on the relevant filesystem to determine whether capacity may contribute.
- Avoid unnecessary storage-cleanup procedures when storage is not implicated.

8. Safe Process Management

- Prefer controlled test processes for training and validation.
- Terminate only known, safe, and authorized processes.
- Avoid killing system services or unknown processes.

9. Evidence Collection

- Record diagnostic results and relevant observations.
- Keep evidence in the designated project Evidence directory.
- Redact sensitive information before sharing screenshots or reports.

10. Escalation Criteria

- Escalate persistent performance problems that cannot be safely resolved.
- Escalate suspected hardware, kernel, application, or infrastructure problems.

11. Best Practices

- Diagnose before changing the system.
- Correlate multiple performance indicators.
- Make the smallest authorized change necessary.
- Verify the result after remediation.
- Document the complete troubleshooting path.